

Draft Fans (ID / FD / PA) - O&M Manual

Axial/centrifugal draft fans; 2x50% per service

Field	Value
Document No.	FAN-OM-3050
Revision	Rev 3.1
Equipment OEM	(synthetic; BHEL class)
Applicable standard	ISO 10816 / IEC / CEA norms
Classification	O&M reference - SYNTHETIC (Ballast demo)
Owner	Plant O&M / Reliability

Disclaimer: This is a synthetic document generated for the Ballast demo. Values are grounded in public standards and typical industry practice but do not represent any specific OEM's proprietary manual.

1. Overview

Induced-draft (ID), forced-draft (FD) and primary-air (PA) fans establish furnace draft and combustion/transport air. Configured 2x50%; loss of one fan derates the unit (Criticality 'A' for ID/FD/PA).

2. Vibration Limits (ISO 10816)

Zone	Vel. RMS (support-A)	Vel. RMS (support-B)	Action
Zone A (new/good)	<= 2.3	<= 3.5	Newly commissioned; ideal
Zone B (acceptable)	2.3 - 4.5	3.5 - 7.1	Unrestricted long-term operation
Zone C (alarm)	4.5 - 7.1	7.1 - 11.0	Short-term only; plan corrective action
Zone D (danger)	> 7.1	> 11.0	Risk of damage; trip / immediate action

Point	Normal	Alarm	Trip
Fan bearing vibration	Zone A/B	4.5 mm/s	7.1 mm/s
Bearing metal temp	50 - 75 degC	90 degC	100 degC

3. Service Intervals & Failure Modes

Task	Interval
Bearing regreasing	2,000 hrs
Bearing replacement	On trend / 20,000 hrs
Impeller erosion inspection (ID)	Annual
Blade-pitch actuator check	Quarterly

- Imbalance (ash deposit on ID impeller) - 1x vibration rise; clean/rebalance.
- Bearing fatigue - broadband + bearing-frequency vibration; replace.
- Duct/damper issues - flow instability, draft loss.